

# Beatty Secondary School

## NX 3023 Research Proposal

### Group 1

Group Members: Louis, Lucas, Ganith, Meng Yeoh & Darius

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| <b>Research Question</b> | <b>Does listening to classical (piano) music affect peoples' memory?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>List of variables</b> | <p>Independent: The type of music</p> <p>Dependent: The memory performance (The result of the test)</p> <p>Constant: Room temperature / volume of music / number of words; etc.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Hypothesis</b>        | People who listen to classical piano music will be able to remember <b>1≥ words more</b> than without music.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Test Design</b>       | <p><b>Who?</b> Principal and Vice Principals-&gt; (1 principal and 2 vice principals)<br/> Teachers-&gt; (11 teachers)<br/> Seniors-&gt; (8 boys and 8 girls)<br/> Juniors -&gt; (Backup; to replace teachers, seniors, and principals if they are absent)</p> <p><b>What?</b></p> <ol style="list-style-type: none"> <li>1. 2 different lists of words for two sections of the test paper, Section A and Section B. (Circle 15 correct words out of the 50 total words in the word list. 15 correct, 35 incorrect.)</li> <li>2. 1 min to memorise</li> <li>3. 5 min timer to test</li> <li>4. Time &amp; Date of the test will be decided on a suitable time for the subjects. (Different time and date for different individuals)</li> <li>5. Place will be in Comp Lab 1/2</li> <li>6. Repeat the same process but when memorising the words, listen to the classical (piano) music</li> <li>7. Headphones will be provided (but best to bring on their own for hygiene/comfortability purposes)</li> </ol> <p><b>How?</b><br/> Giving a memory test (remember 15 words and circle them in a word list)</p> <p><b>Key findings:</b> When we did a testing on our own on 28/5/25,</p> |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>We found that the timing was too long to memorise the words. The average test scores were high, 13/15. So, made some changes. We shortened the time for memorising from 5min to 1min, and as well as making the difficulty of the words to be harder by changing the format the words such that correct and wrong words are not cramped together. (Correct words are too close to each other and not separated far away from each other).</p> |
| <b>Planning of Data Table</b>      | Based on teacher's and student's scores, with and without music (recording in bar graph)                                                                                                                                                                                                                                                                                                                                                         |
| <b>Analyse data for conclusion</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### **Classical music we are using**

<https://www.youtube.com/watch?v=rdoq4yi9cV0&t=804s>

4 pieces of classical songs

### **People we are planning to test this out on**

#### *Teachers*

1. Mr Goh—subject 001
2. Ms Ling —subject 002
3. Ms Ain —subject 003
4. Mrs Seah— subject 004
5. Mr Ganesh—subject 005
6. Mr Fadzrun — subject 006
7. Ms Kang—subject 007
8. Mr Ben— subject 008
9. Mr Jack—subject 009

#### *Students*

##### **Male**

1. Alerick—subject 010
2. Poova—subject 011
3. Gabriel—subject 012
4. Jayden (Wushu) —subject 017
5. Christian— subject 018

6. Lucas Tan— subject 019
7. Zale—subject 020
8. Lucas Cheah—subject 021

### **Female**

1. Mindy— subject 022
2. Anisha—subject 023
3. Jia Xuan—subject 024
4. Kayle—subject 025
5. Xing Ying —subject 026
6. Yi Qing— subject 027
7. Rinoa—subject 028
8. Emma—subject 029

### **List down the sources of your research**

Impact of music learning on students' psychological development with mediating role of self-efficiency and self-esteem: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11371213/>

Our inspiration for the test (false memories):

[https://faculty.washington.edu/chudler/chmemory.html?utm\\_source=chatgpt.com](https://faculty.washington.edu/chudler/chmemory.html?utm_source=chatgpt.com)

How music Affects Memory and concentration:

<https://www.levinemusic.org/about/news/how-music-affects-memory-and-concentration/>

How can music improve your memory and concentration:

<https://www.twelvetonemusicschool.com/how-music-can-improve-your-memory-and-concentration/>

Our proof music helps with memorising: <https://today.usc.edu/does-music-unlock-memory/>

### **Word list 1 answers (with music)**

Balloon

Clock

Blue

Heap

Candle

Guitar

Bicycle  
Cloud  
Window  
Forest  
Rock  
Banana  
Computer  
Castle  
Feather

Test file(list 1): [https://beattyss-my.sharepoint.com/:w:/g/personal/phang\\_you\\_xi\\_darius\\_btyss\\_moe\\_edu\\_sg/ERcqlNlDK9BGnAEdwH6aapsB1-ZRfp-PUBkabe1yse0BCA?e=Q7XEK0](https://beattyss-my.sharepoint.com/:w:/g/personal/phang_you_xi_darius_btyss_moe_edu_sg/ERcqlNlDK9BGnAEdwH6aapsB1-ZRfp-PUBkabe1yse0BCA?e=Q7XEK0)

Word list 2 answers (without music)

Ticket  
Star  
Lake  
Circle  
Green  
Valley  
Page  
Lamp  
Trumpet  
Scooter  
Air  
Door  
Garden  
Ruby  
Ball

Test file(list 2): [https://beattyss-my.sharepoint.com/:w:/g/personal/phang\\_you\\_xi\\_darius\\_btyss\\_moe\\_edu\\_sg/EU3pVKc3xF5Mn\\_PaiuvTb-YBlDCjjig9EF9ak0VX\\_Y-xTg?e=OpnC2U](https://beattyss-my.sharepoint.com/:w:/g/personal/phang_you_xi_darius_btyss_moe_edu_sg/EU3pVKc3xF5Mn_PaiuvTb-YBlDCjjig9EF9ak0VX_Y-xTg?e=OpnC2U)

**Our roles**

- 1) Darius & Lucas (Data Collector and analysis)**
- 2) Ganith & Meng Yeoh (invigilator)**
- 3) Louis (timekeeper)**

**Ganith- marker for section A (with music)**  
**Louis- marker for section B (without music)**

## Timeline and script

### Phase 1

Once the people have arrived, Louis and Lucas will present while Darius will set up the headsets while ganith and Meng Yeoh will set up the classical music on the laptops.

If laptops do not work, students can use theirs phones and earpieces to conduct the test.

**Intro script:** Louis: hey guys, welcome to our nexus a star programme. Thanks participating in our project on whether classical music affects one's memory. Without further ado, I shall let Lucas brief y'all on what to do.

LUCAS: so our test have 2 parts. One part with music and the other part without. I will be flashing you the words that u will need to memorise for 1 minute and then after that u will be given the paper which u will need to circle out the words. DO NOT CIRCLE EVERYTHING AND CIRCLE WHATS GIVEN.

Louis: so for the paper with music there will be 15 words u will need to circle while for the paper without music u will need to circle 14 words. You may now put on ur headset and play the classical music as I show u the list of words for a minute. (Pls watch them as they might not be on the right page)